

Grid Trading in Cryptocurrency Markets: A Critical Evaluation of Classical, Dynamic, and Delta-Neutral Implementations

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Abstract

Grid trading is one of the most widely marketed algorithmic strategies on retail cryptocurrency exchanges, typically presented through screenshots of positive profit-and-loss curves with little disclosure of deployed capital, holding period, or fee tier. This note examines whether grid trading generates an independent trading edge or whether its apparent profitability is better explained by market exposure, execution economics, and market-making infrastructure. We review three mechanically distinct implementations discussed in the recent literature — classical grid trading, Dynamic Grid Trading (DGT), and delta-neutral grid market making — and evaluate the evidentiary basis behind each. For classical grid trading, we rely on an existing analytical result showing zero expected profit under an explicitly stated symmetric random-walk model before costs. For Dynamic Grid Trading, we show that the reported outperformance over buy-and-hold is attributable in part to retained directional exposure over a period of substantial cumulative price appreciation, a point the source paper’s own authors acknowledge. For delta-neutral grid market making, we identify an unresolved capital-accounting inconsistency and an incomplete fill model in the primary source, together with citation inconsistencies between its in-text attributions and its bibliography, and we supplement this with an original unit-economics illustration showing that the sign of a representative spot-leg cycle’s economics flips from positive to negative when moving from the paper’s assumed maker rebate to a standard retail maker fee. Taken together, the evidence indicates that grid trading is not a single strategy with a single source of return: as implementations move from a simple grid toward market-neutral market making, the economic source of any edge shifts away from the grid structure itself and toward fee schedules, execution quality, adverse-selection control, and market microstructure. We do not conduct an independent empirical replication of any of the three implementations; our contribution is a structured, source-verified synthesis of the existing evidence, a citation-consistency check on one of the primary sources, and a self-contained fee-sensitivity illustration.

Keywords: grid trading, algorithmic trading, cryptocurrency markets, delta-neutral strategies, market making, market microstructure, transaction costs, Binance

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1. Introduction

Grid trading bots are marketed on cryptocurrency exchange platforms as a mechanical, direction-agnostic way to profit from price oscillation: place a ladder of buy and sell orders across a price range and collect the spacing between filled levels as price moves up and down inside that range. Retail marketplaces commonly display this strategy’s track record as an absolute profit-and-loss figure — for example, “+\$5,000 in a day” — without disclosing the capital deployed, the length of the period, or the fee tier under which the result was achieved. Because the mechanism is publicly documented and available to any account on the exchange, the relevant research question is not whether the mechanism itself is legitimate, but whether it constitutes an independent source of trading edge once market exposure and execution economics are properly accounted for.

This note addresses that question for three implementations discussed in the recent literature: classical (static) grid trading, Dynamic Grid Trading (Chen, Chen, and Jang, 2025), and delta-neutral grid market making (Nguyen and Bui, 2026). For each, we ask the same question this broader research series applies to every strategy under review: how much of the reported result belongs to the mechanism itself, and how much belongs to the conditions — market regime, fee structure, execution assumptions — under which it was measured.

Section 2 describes the mechanics of a classical grid. Section 3 reviews the analytical zero-expectation result for classical grid trading under a random-walk model. Section 4 examines Dynamic Grid Trading and the source of its reported outperformance. Section 5 examines delta-neutral grid market making, including a capital-accounting inconsistency, an incomplete fill model, and a citation-consistency issue identified in the primary source, together with a broader methodological point on single-sample backtest validation drawn from an independent line of literature. Section 6 presents an original fee-sensitivity illustration built on the delta-neutral paper’s own stated parameters. Section 7 discusses what additional sophistication in a grid-based system actually purchases in economic terms. Section 8 addresses the interpretation of retail marketplace profit-and-loss disclosures. Section 9 states the benchmark standard this series applies before any strategy is considered to have a usable edge. Section 10 concludes.

2. Mechanics of a Classical Grid

A classical grid strategy sets a price range, places a ladder of price levels inside it, and posts a buy order at every level below the current price and a sell order at every level above it. As price oscillates within the range, filled buy orders are paired with filled sell orders at the level above, and the strategy earns the spacing between grid levels on each completed round trip.

The strategy’s viability is conditioned on price remaining inside the defined range. If price falls below the lower bound, the position accumulated on the way down is left unhedged against further downside, with the strategy either holding the asset with

no further grid participation or liquidating at a loss, depending on configuration. If price rises above the upper bound, the entire position has already been sold and the strategy no longer participates in further appreciation. In either direction, a classical grid’s return profile depends on price moving sideways rather than trending.

3. The Zero-Expectation Result for Classical Grid Trading

Chen, Chen, and Jang (2025) formalize the classical grid’s return profile under an explicit random-walk model: price moves up or down by a fixed step in each period with equal probability, transaction costs are set to zero, and the position is fully liquidated once price breaches the lower bound of the range. Under these stated assumptions, the authors derive analytically — rather than observe empirically on a single data sample — that the expected profit of a classical grid strategy is zero before costs.

This result is a proof under one explicitly stated model, not a general claim about grid trading’s expected value under arbitrary market conditions. It is nonetheless directly relevant to how retail marketplaces present the strategy: a single realized profit-and-loss curve over one historical period is a draw from a distribution, not evidence of the distribution’s mean, and a random-walk model with zero expected value before costs is entirely consistent with individual realized paths showing large positive or negative outcomes.

4. Dynamic Grid Trading and the Source of Its Reported Outperformance

Chen, Chen, and Jang (2025) also examine a variant they term Dynamic Grid Trading (DGT), which differs from the classical implementation in two respects: rather than halting at a range boundary, the grid recenters on the current price, and rather than liquidating on a downside breach, the strategy retains the asset. Tested on data from January 2021 to July 2024, DGT’s reported internal rate of return exceeds buy-and-hold for Bitcoin, with a comparatively lower maximum drawdown; for Ethereum, the return advantage over buy-and-hold is considerably smaller, while the drawdown advantage persists. The effect is therefore not uniform across the two assets tested.

The authors state the likely source of the result directly, attributing the strong performance largely to the cumulative rise in cryptocurrency prices over the sample period. The period is not uniformly bullish — it spans the 2022 downturn — but the net cumulative appreciation over the full window is, by the authors’ own account, a material driver of the reported outperformance. The mechanical interpretation is that a classical grid harvests price oscillation within a range, whereas DGT additionally retains directional exposure to where price ultimately travels; accordingly, DGT should not be characterized as market-neutral alpha, irrespective of the smoothness of its draw-down profile.

5. Delta-Neutral Grid Market Making

Nguyen and Bui (2026) propose a structurally different mechanism: grid quoting on the spot market combined with directional hedging via futures, intended to isolate

profit from bid-ask spread capture — in the manner of conventional market making — rather than from asset appreciation.

5.1 Reported results and profit decomposition

The authors report a backtest on Gate.io BTCUSDT over 52 trading days (November–December 2025), decomposing total profit and loss into three components: spread and grid capture (approximately 76% of total P&L), funding fees (approximately 18%), and maker rebates (approximately 6%). The headline reported results are a total return of approximately 37% over the 52-day window, a daily Sharpe ratio exceeding 3.0, and a maximum drawdown below 8% of net asset value.

One point of scope requires clarification: while the paper’s abstract and architecture description reference Binance as the exemplifying venue, the empirical backtest underlying the reported figures was conducted on Gate.io. This is not represented by the authors as a Binance validation, but it should be stated plainly for readers evaluating applicability: the architecture is proposed for Binance-like spot/futures market structures, and the empirical evidence supporting the headline figures comes from a different exchange.

5.2 An unresolved capital-accounting inconsistency

The paper states an initial capital allocation of 100 USDT-equivalent, split evenly between spot and futures legs (50 USDT each). The stated base order size per grid level, however, is 0.01 BTC across five levels per side, with a V-shaped sizing scheme scaling individual levels up to 0.035 BTC. At the price levels under discussion, a single base order of 0.01 BTC carries a notional value on the order of a thousand US dollars — substantially larger than the stated 50 USDT allocated to that leg, before accounting for the additional grid levels. The visible text of the paper does not present a leverage or margin-accounting explanation that reconciles the stated capital base with the stated order sizing. We treat this as an unresolved inconsistency rather than a demonstrated error; it does, however, mean that the reported 37% return relative to a 100 USDT net asset value cannot be treated as an independently verified, reproducible figure absent clarification of the margin treatment or a transaction-level replication.

5.3 A citation-consistency issue

Independent of the empirical results, we identify an inconsistency between two in-text citations in the paper’s Related Work section and the corresponding bibliography entries. Leippold, Wang, and Zhou are described in-text as documenting the sensitivity of retail grid strategy profitability to grid range calibration on Binance, while the bibliography entry under the same citation is titled “Machine Learning in the Chinese Stock Market,” published in the *Journal of Financial Economics*. A parallel pattern appears for Brunnermeier, Sockin, and Xiong, cited in-text for empirical findings on the autocorrelation structure of cryptocurrency perpetual funding rates, while the corresponding bibliography entry is titled “China’s Model of Managing the Financial System,” published in the *Review of Economic Studies*. We do not interpret this as

an indication of fabricated results; a bibliography-compilation error is a plausible and more likely explanation. We report it because at least two in-text citations in a section supporting the paper’s empirical framing do not appear to correspond to the works listed under those citations in the bibliography, which is a relevant consideration when weighing the paper’s empirical claims.

5.4 The fill-model problem

The authors disclose several limitations of their own backtest: a 52-day sample with no validation over an extended bear market, a fill model that does not account for partial fills, no explicit modeling of queue position, and heuristic rather than data-derived weights for toxic-flow detection.

The queue-position limitation warrants elaboration, as it is a load-bearing assumption rather than a minor caveat. In a real limit order book, the fact that market price touches or crosses a resting order’s price level does not guarantee that a specific order at that level is filled: other orders may sit ahead of it in the price-time priority queue, and the order in question fills only after those orders are exhausted, if at all before price reverses. For a strategy whose reported backtest comprises approximately 14,000 completed cycles, treating each theoretical fill opportunity as realized is a substantial assumption, and it is not one that the backtest itself tests. For this reason, the reported Sharpe ratio, drawdown, and total return figures are best read as outcomes of one specific backtest under stated modeling assumptions, rather than as established properties of the strategy under live execution.

5.5 A methodological note on single-sample validation

A broader methodological consideration, not specific to this paper, is relevant here. Gort, Liu, Gao, Chen, and Wang (2022/2023) show, in the context of cryptocurrency markets, that a conventional walk-forward validation with a single train/test split is prone to overfitting, and that k-fold cross-validation — a common alternative — relies on an independent-and-identically-distributed assumption that does not hold for financial time series. We have not computed a formal backtest-overfitting probability for Nguyen and Bui’s specific implementation; that would constitute a separate study. The applicable general point is that a single 52-day sample on one exchange does not, on its own, establish robustness, consistent with the broader backtesting-methodology literature on crypto markets.

6. A Fee-Sensitivity Illustration

To make the practical stakes of fee structure concrete, we construct a simplified unit-economics illustration of a single spot-leg round trip, using the spread parameter stated in Nguyen and Bui’s paper. This is explicitly not a reproduction of the authors’ backtest and does not model the futures hedging leg, hedging costs, adverse selection, or slippage; its sole purpose is to show how the sign of a representative single-cycle calculation responds to a change in fee assumptions.

Holding the round-trip spread fixed at $\delta = 0.10\%$ (the value used in the source paper) and applying a fee to both sides of the cycle, three fee assumptions yield three

different outcomes. Under the paper’s own assumed maker rebate of -0.025% per side, the cycle nets $+0.15\%$ (spread of $+0.10\%$ less a fee of -0.05% across both sides, where the negative fee is a rebate paid to the trader). Under a retail maker rate with a 25% BNB fee discount, $+0.075\%$ per side, the cycle nets -0.05% (spread of $+0.10\%$ less fees of 0.15% across both sides). Under the standard retail maker rate with no discount, $+0.10\%$ per side, the cycle nets -0.10% (spread of $+0.10\%$ less fees of 0.20% across both sides).

The approximately 14,000 completed cycles reported in the source paper’s backtest cannot be applied directly to this per-cycle arithmetic to derive a cumulative return, because each cycle trades a different share of notional and capital carries over between cycles — compounded further by the capital-accounting ambiguity noted in Section 5.2. The paper’s headline reported return of approximately 37% over 52 days is a separate, independently reported figure that this single-cycle calculation can neither reproduce nor refute. What the calculation does show, and what can be stated with confidence, is that moving from the paper’s assumed negative fee (a rebate) to a standard positive retail fee flips the sign of the per-cycle unit economics. This does not establish that the full strategy — inclusive of its hedge — is unprofitable at retail fee tiers, which would require a complete transaction-level replication; it does establish that the paper’s reported result does not transfer to a retail fee tier without independent recalculation.

7. What Additional Sophistication Actually Purchases

It is useful to enumerate what the Nguyen and Bui architecture adds beyond the grid mechanism itself: volatility-adaptive level spacing, a dynamically recentered grid using an exponential moving average, trend-based quote skewing, order-flow imbalance measured over a trailing 60-second window, funding-rate accounting, inventory-aware quoting, regime detection via the Average Directional Index, realized spread measurement, asymmetric fill modeling between the two sides of the book, tracking of adverse price movement in the five seconds following a fill, toxic-flow detection, circuit breakers, and cross-venue spot/futures hedging.

At this level of construction, the system is a full market-making architecture with an inventory-management layer, a microstructure-analysis layer, and an execution/risk layer, in which the grid’s order-placement logic is a mechanism rather than an independent source of return. The general pattern is that as an implementation moves toward a more sophisticated, market-neutral design, less of its return can be attributed to the grid structure itself, and more of it is properly attributed to spread capture, fee schedule, execution quality, adverse-selection control, inventory management, and market microstructure — domains in which the grid is, at most, one component among several.

8. Interpreting Retail Marketplace Profit-and-Loss Disclosures

Retail bot marketplaces commonly display absolute profit figures — for example, a stated gain over a stated period — without consistently disclosing deployed capital, return as a percentage of capital, the length of the measurement period, or the applicable fee tier. Some marketplace interfaces do report ROI or invested capital along-

side the P&L figure; where this information is absent or not prominently displayed, an absolute P&L number in isolation provides no basis for assessing reproducibility. An identical absolute profit figure realized on a small deposit versus a substantially larger one represents two fundamentally different outcomes that are indistinguishable from the displayed figure alone.

9. The Relevant Benchmark Standard

Even where a favorable fee tier is available and a delta-neutral grid market-making implementation produces a positive return, an additional condition must be satisfied before the result constitutes a usable edge: the return must exceed what could be earned from the best available alternative use of the same capital, at comparable risk and effort. This is the same benchmark standard this broader research series applies uniformly across strategies, including carry-based strategies examined elsewhere in the series. This note does not conduct a quantitative benchmark test for any grid implementation; the standard is stated here as the applicable decision framework, not as an empirical result of this note.

10. Conclusion

Grid trading is not a single strategy with a single source of return; it is a family of at least three mechanically distinct implementations with different evidentiary support. For classical grid trading, an explicit analytical result under a stated symmetric random-walk model shows zero expected value before costs, with transaction costs further worsening the economics. For Dynamic Grid Trading, the reported outperformance over buy-and-hold is attributable in part to retained directional exposure realized over a period of substantial net price appreciation, a point acknowledged by the source paper's own authors. For delta-neutral grid market making, the primary source carries an unresolved capital-accounting inconsistency, an incomplete fill model that does not account for queue position, and a citation-consistency issue between its in-text attributions and its bibliography; accordingly, its reported results are best characterized as outcomes of one specific backtest under stated assumptions rather than as established properties of the strategy. An original fee-sensitivity illustration presented here shows that the sign of a representative single-cycle unit-economics calculation, built on the source paper's own parameters, reverses when moving from the paper's assumed maker rebate to a standard retail maker fee.

The general finding is that grid, as a mechanism, is not itself an independent source of trading edge. As implementations progress from a simple grid toward a robust, market-neutral design, the economic source of any return shifts away from the grid structure and toward market microstructure, execution quality, and fee economics. This conclusion should not be overstated in either direction: we have not conducted an independent, transaction-level replication of Dynamic Grid Trading or of Nguyen and Bui's delta-neutral architecture, and the evidence assembled here — an analytical result under stated assumptions, one historical backtest, one proposed and partially tested architecture, and one fee-sensitivity illustration — does not support a categorical claim that retail grid trading is proven unprofitable, nor that delta-neutral grid market making cannot work for a retail participant. The accurate statement is nar-

rower: the published evidence reviewed here does not establish that the reported results transfer, without further verification, to standard retail execution and fee conditions.

Data and Code Availability

This note is part of an ongoing, publicly documented research series, *Searching for Edge*, which applies the same evidentiary standard — literature review, replication where feasible, net-of-costs adjustment, and significance testing — to a set of systematic cryptocurrency trading strategies. The series, including this note and related material, is available at: <https://github.com/Silent47boryara/searching-for-edge>.

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